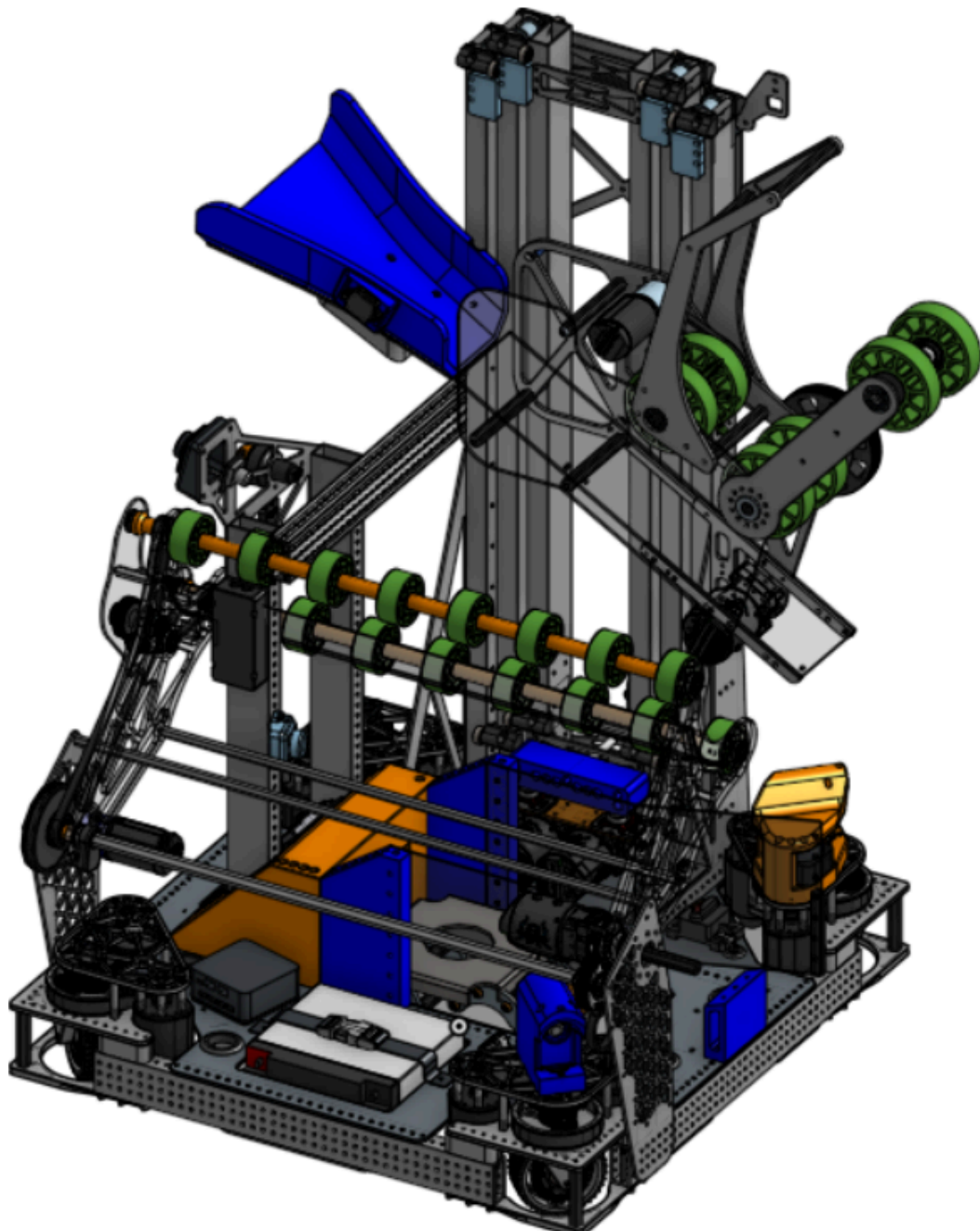


Sharkira

2024-2025

ENGINEERING PROGRESS AND DEVELOPMENT

FRC TEAM 1073: THE FORCE TEAM

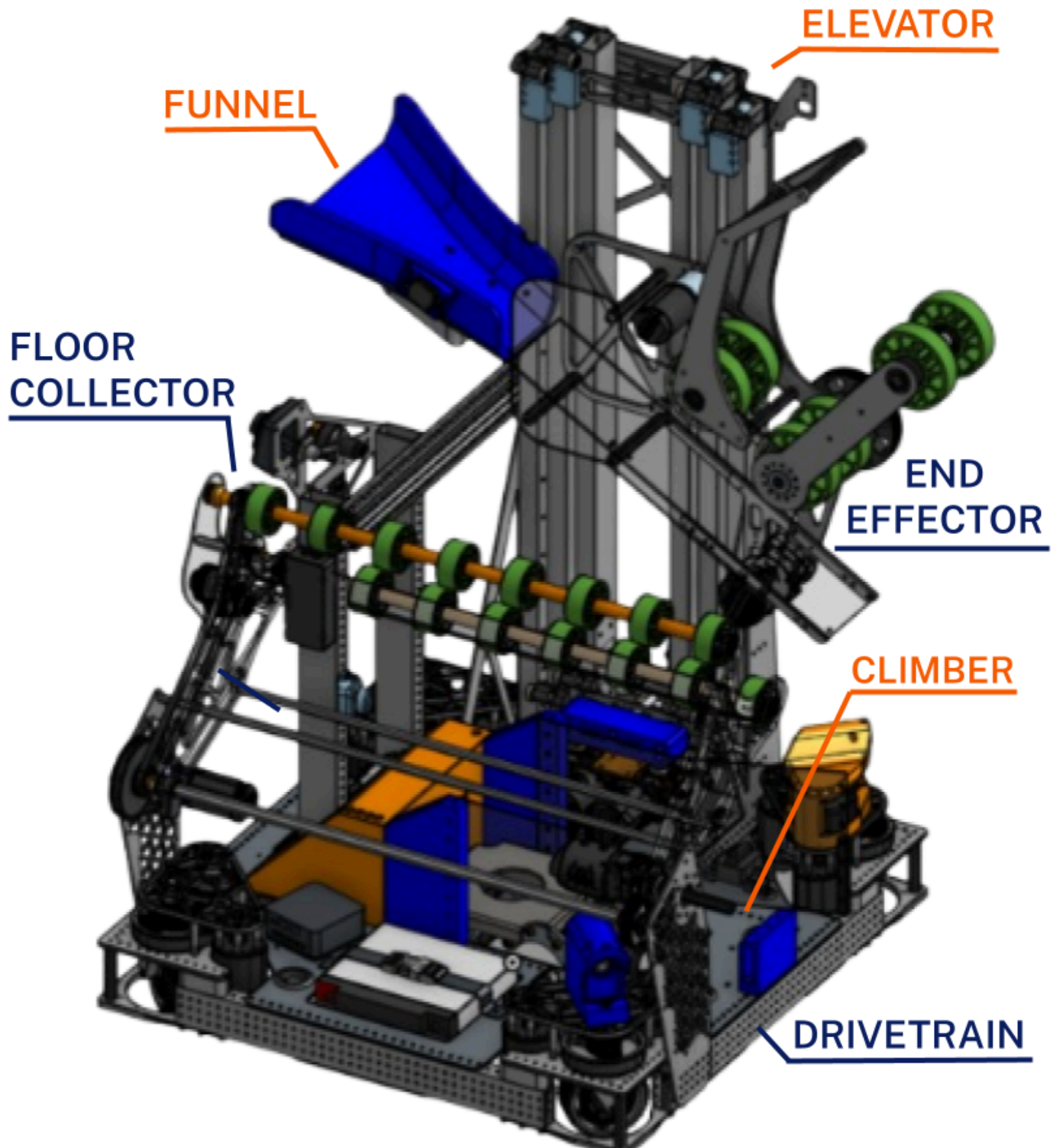


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FULL ROBOT CAD & MECHANISM LANGUAGE

SHARKIRA



PLANNING AND SCHEDULE

2024-2025

Sharkira

REQUIREMENTS

Team Strategy

AUTONOMOUS

*Start center relative to the Driver Station

*Preload Coral

- | | |
|------------------------------------|------|
| - LEAVE starting line | 3pts |
| - Score preload CORAL in REEF L4 | 7pts |
| - Collect CORAL from CORAL STATION | — |
| - Score CORAL in REEF L4 | 7pts |
| - Traverse to CORAL STATION | — |

TELE-OPERATED (Cycle)

*Cycle time = time to drive to, collect, manipulate, and score Cargo

- | | |
|--------------------------------|------|
| - Score L1-L4 | |
| - L1 | 2pts |
| - L2 | 3pts |
| - L3 | 4pts |
| - L4 | 5pts |
| - Score Algae in the Processor | 6pts |
| - Be able to play defense | — |

ENDGAME

- | | |
|-------------------|-------|
| - Climb Deep Cage | 12pts |
|-------------------|-------|

PRIORITIES

1. Endgame (Deep Cage)
2. Coral Scoring
3. Algae Scoring
4. Reduce Cycle Time

Functional Requirements

The first set of requirements are intended to be strategy-based, in order to guide our design without pre-designing our robot. In past years we struggled with focusing too much on the robot detail instead of what the strategy demanded of the robot in terms of performance.

Specific Requirements

We determined our specific requirements further along in the design process so we would understand the amendments and clarifications that needed to be made. We then had our Software and Electromechanical subgroups add their own needs, requirements, and comments, and put those to a vote.

Please see 1073's Requirements/Verifications Binder for more details on our requirements process.

DEVELOPMENTAL PLAN & DESIGN REVIEWS:

Roadmap to Worlds

In the offseason, our student leadership team developed a plan entitled, “Roadmap to Worlds”, in an effort to streamline our design, manufacturing, assembly, and integration process to increase the amount of software development time.

- 4 January - Strategic Development**
 - Game manual analysis & quiz
 - Brainstorming, cost vs. benefit analysis
 - Strategic proposal decision & outline
- 5 January - Functional Requirements**
 - Prototyping groups begin on assigned mechanisms
- 6 January - Drivetrain CAD Complete**
- 8 January - Design Review 1**
- 15 January - Design Review 2**
- 18 January - Design Review 3**
- 21 January - Wiring and Testing Drivetrain**
- 26 January - Mechanisms Built**
- 2 February - Mechanisms Integrated, Initial Testing**
- 3 February - Robot Assembly Complete**
- 12 February - Testing, Development, Drivetrain Practice, Iterating**
- 15 February - Week 0 Competition**
- 16 February - Drivetrain Practice, PR Practice, Awards Development, Autonomous**
- 6 March - Competition 1 Pack Night**

Though the team considers this an ambitious schedule, and we often fall behind due to weather or manufacturing constraints out of our control, having a system in place sets us up for success. We had a fully functional robot, were in the midst of award preparation for various submissions and presentations, practiced regularly with drivetrain, and had a strong autonomous between February 14th and March 1st.

This presentation featured our design review process, decision making flow, priorities for awards at DCMF and Worlds, and other competition plans. See below (and on the next page) for sample slides.

10

Strategy Creation Methods

EVERYONE who passes the rules quiz will be participating and contributing to this process.

1. Divide into groups with multi-subgroup influence and one mentor
2. Each group analyze game and come up with complete strategy
3. Find commonalities in all strategies and determine these to be highest priority along with those of the highest point/rank point contribution
4. Discuss all tradeoffs and acknowledge limitations as a result of our decisions

*Vote on tradeoffs

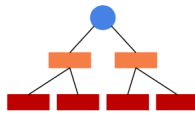
*Vote on final decisions within subsections of the strategy

*Vote on the final strategy/determine need for changes

*** $\frac{2}{3}$ Of attending students needed to agree on the strategy before final decision from the outlined group**

→ If we follow a “flowchart” idea, then tradeoffs are apparent and once we decide to favor one idea over another, we decide to favor all related ideas

09



Strategic Development and Process

e.g. 2023 Game: We want a robot that can score in both of the top nodes

- Endgame
 - Climb
- Auto
 - 2 Game Pieces
 - Collect
- Tele
 - Apriltag Alignment
 - Top Node Scoring
- General
 - Pick up off ground and substations

The goal is to develop a generic list of what a robot will do during a match.

WHAT not HOW

This strategy will be SET IN STONE and not change except for extenuating circumstances.

FRC 1073

Design Review Process

At every level of our design review process, we keep the principles of integration in mind. When selecting each mechanism/designing each mechanism, we designate and consider the ways it will interact with the remainder of the robot.

- Design Review 1: Intended to be a feasibility pass with mentors to determine whether or not to waste time and/or resources on an idea. Produces a document of proposed ideas, changes, questions and specs.
- Design Review 2: Intended to be a means by which to collaborate with software and strategy with a functioning prototype and partial CAD design to display. Produces a document of proposed ideas, changes, questions, and specs, slims down the options for each mechanism.
- Design Review 3: Final pass on each mechanism to ensure it falls within requirements, display a final CAD, begin BOM creation process, and begin a COTS list before finalization and fabrication. Produces needed changes before fabrication, selects the one mechanism to fabricate.

PROJECT SCHEDULES

GANTT CHART

The team utilized this method to keep track of our developmental process, create time expectations for our subgroups, and to ensure none of our students were inactive when we could be making progress toward our goals.

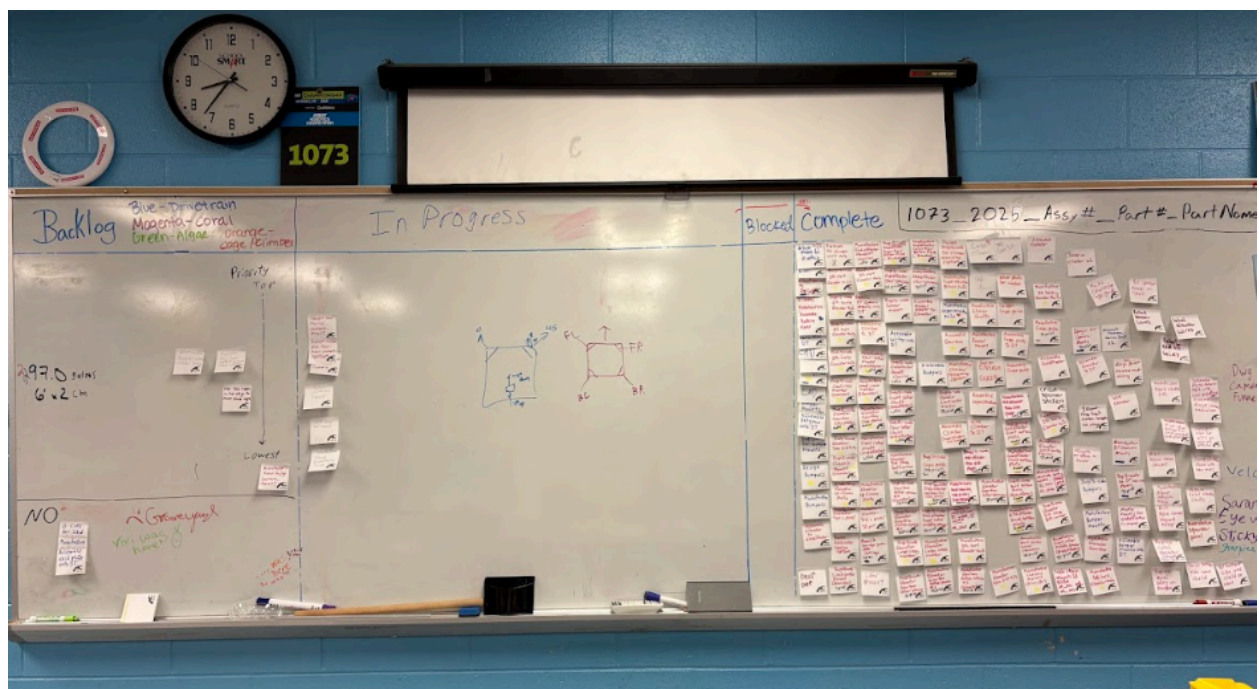
In the first week of Kickoff, our student leads made a list of what needed to be done throughout the season, and assigned students to tasks based on their role in their subgroup (i.e. Climber Lead).

Throughout the meeting, our CEOs would update the progress and work to adjust expected completion dates to match our current track, depending on what did or did not get accomplished on that day. The subsequent task due dates would be cascaded accordingly.

Updated as of: 3/6/2025							
WBS	Task Name	Start Date	End Date	Assigned To	SUBGROUP	% Done	Stage
A MILESTONES							
A1	Design Review 1	1/8/2025	1/8/2025	EM	MILESTONE	100%	Done
A2	Design Review 2	1/15/2025	1/15/2025	EM	MILESTONE	100%	Done
A3	Drivetrain Assembly Complete	1/18/2025	1/18/2025	EM	MILESTONE	85%	In Progress
A4	Design Review 3	1/22/2025	1/22/2025	EM	MILESTONE	100%	Done
A5	Mechanism Built & Programmed	2/1/2025	2/1/2025	EM, SW	MILESTONE	100%	Done
A6	Robot Assembly Complete	2/4/2025	2/4/2025	EM	MILESTONE	99.9999%	In Progress
A7	Week 0	2/14/2025	2/15/2025	1073	MILESTONE	100%	Done
A8	Week 2 Waterbury	3/7/2025	3/9/2025	1073	MILESTONE	0%	Backlog
A9	Week 4 WPI	3/21/2025	3/23/2025	1073	MILESTONE	0%	Backlog
A10	Week 5 UVM	3/28/2025	3/30/2025	1073	MILESTONE	0%	Backlog
A11	DCMP	4/2/2025	4/5/2025	1073	MILESTONE	0%	Backlog
A12	Worlds	4/15/2025	4/20/2025	1073	MILESTONE	0%	Backlog
1	Business & Competition Strategy					49.58850846%	
1.1	Field Taping	1/4/2025	1/8/2025	Thara, Benton, Phea (BACS)	BACS	100%	Done
1.2	Robot Name Selection	2/4/2025	2/5/2025	1073	BACS	100%	Done
1.3	Bumpers		2/14/2025	Thara, Sapph	BACS	100%	In Progress
1.4	Drive Team Creation		2/2/2025	BACS, Sapph	BACS	100%	Done
1.5	Impact Award		2/13/2025	Sapph	BACS	878787878%	In Progress
1.6	Merchandise				BACS	75%	In Progress
1.7	PR Training			Xavier, Sapph	BACS	33.33333333%	In Progress
1.8	Quality Assessment			BACS	BACS	4285714286%	In Progress
1.9	Reveal Video		2/27/2025	Sapph, Xavier	BACS	25%	Backlog
1.10	Scouting			Sapph, Head Sco	BACS	60%	Backlog
1.11	Social Media Posts			Phea	BACS	13.33333333%	In Progress
1.12	Sponsor Fulfillment			BACS	BACS	13.33333333%	Backlog
1.13	Transfer Station			Thara	BACS	50%	In Progress
1.14	Woodie Flowers Award		2/12/2025	Sapph, Xavier	BACS	100%	Done
1.15	Website Updates				BACS	0%	Backlog
1.16	Team Spirit			BACS	BACS	27.33333333%	In Progress
1.17	Roadmaps to Worlds Binder		3/5/2025	CEOs	BACS	100%	Done
1.18	Business Binder		3/5/2025	CEOs	BACS		
2	Electromechanical				EM	96.08259587%	
2.1	Drivetrain		2/3/2025	Preston, Gavin, Ya	EM	100%	In Progress
2.2	Climber		2/3/2025	Viraj, Alex D, Vivi	EM	100%	Done
2.3	Coral Mechanism		2/3/2025	Alex T, Roman, Ev	EM	100%	In Progress
2.4	Algae Remover Mechanism		2/19/2025	EM	EM	100%	Backlog

SCRUM

We manage our tasks via Scrum board. We use sticky notes to visibly see what tasks we need to do. It encourages responsibility and helps us see our progression through time. We changed our Scrum board this year. Our written tasks are specific using words such as design, manufacture and assemble. The new style more accurately keeps track of our progress and ensures we properly understand what steps need to be taken to properly complete a part.



ROBOT DESIGN

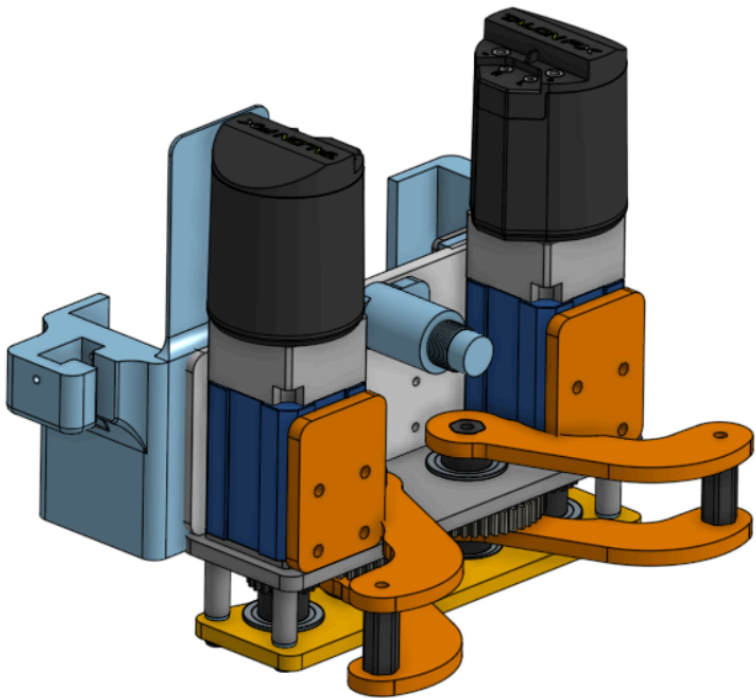
2024-2025

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CLIMBER

2024-2025

DESIGN REVIEWS, ITERATION & FINAL DESIGN



CLIMBER

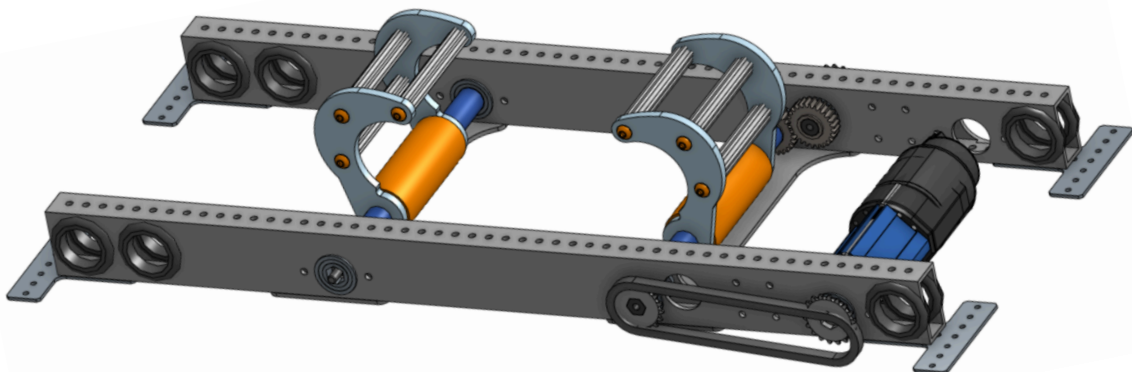
Minimizing the space that the climber occupied while still ensuring fast, reliable climbing was a main priority in this design. We've dedicated the middle of our robot as space for the DEEP CAGE to come in, minimizing the robot's swing due its close proximity to the center of gravity.

VERSION 1:

For V1 (left): the two claws of the climber hook onto the outer bars of the DEEP CAGE and push down, ultimately lifting the robot up. This design exhibited some slip and couldn't necessarily hold the robot "up" long enough.

VERSION 2:

For V2 (below): climber was moved down to the chassis and adapted into a "donut" model, increasing the size of the claws, reducing robot weight, reducing motor count, and eliminating cage slip.



CLIMBER (cont.)

Gearboxes:

- Mounted on the inside of the robot and has a low profile for maximum rigidity and compactness
- Kraken motor input
- Outputs a total gear ratio of 100:1
- Contains a custom profile on the claws to ensure proper grip on the cage
- Manufactured using ¼ inch aluminum
- Max width of 4.25 inches on the claws
- Max length of 4.5 inches

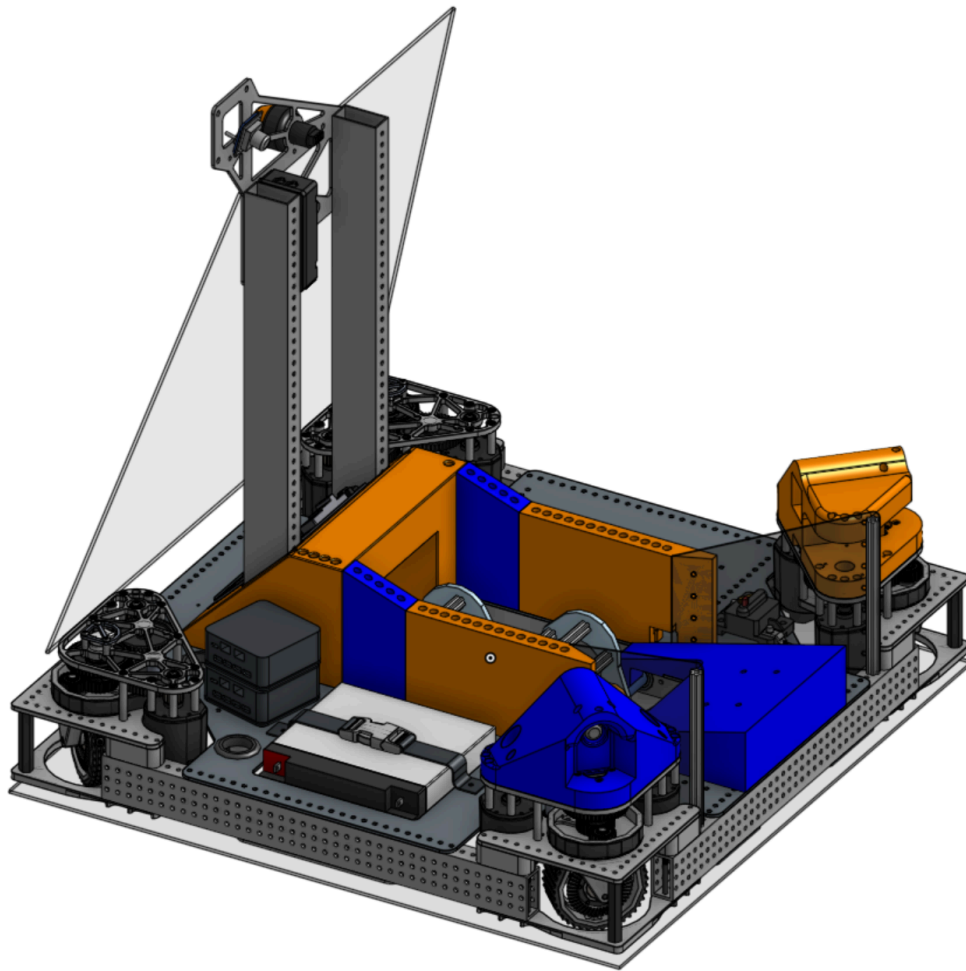
Claws:

- Mounted to the drivetrain frame
- Manufactured using ¼ inch aluminum
- Max width of 4.25 inches
- Max length of 4.5 inches

DRIVETRAIN

2024-2025

DEVELOPMENTAL AND RATIONAL



DRIVETRAIN

Swerve Drive:

- 4 wheel swerve drive
- 4" Mark 4i module's wheels for speed and agility
- 26"x26" milled aluminum frame
- Frame held together with gussets for more accuracy
- Designed in replaceable swerve modules, in case of failure.

DRIVETRAIN (cont.)

Bellypan Layout:

- Thin enough to remain agile based on our center of rotation
- Evenly distributed weight
- Cleanly managed cables
- Center cut out to provide room for the cage during climbing

Gearbox:

- L2 Mark4i Swerve Modules
- 6.75:1 gear ratio
- Exchanged open bearings for closed bearings for robustness

Bumper Mounts:

- Bumper mount screws
- Four L brackets

Camera Mounts:

- Orange bumper mount houses a camera and a LIDAR
- Front facing camera angled downwards at 15 degrees

Super Structure:

- 1x2" tubing
- Custom 1/8" bracket to house RSL and mount cameras

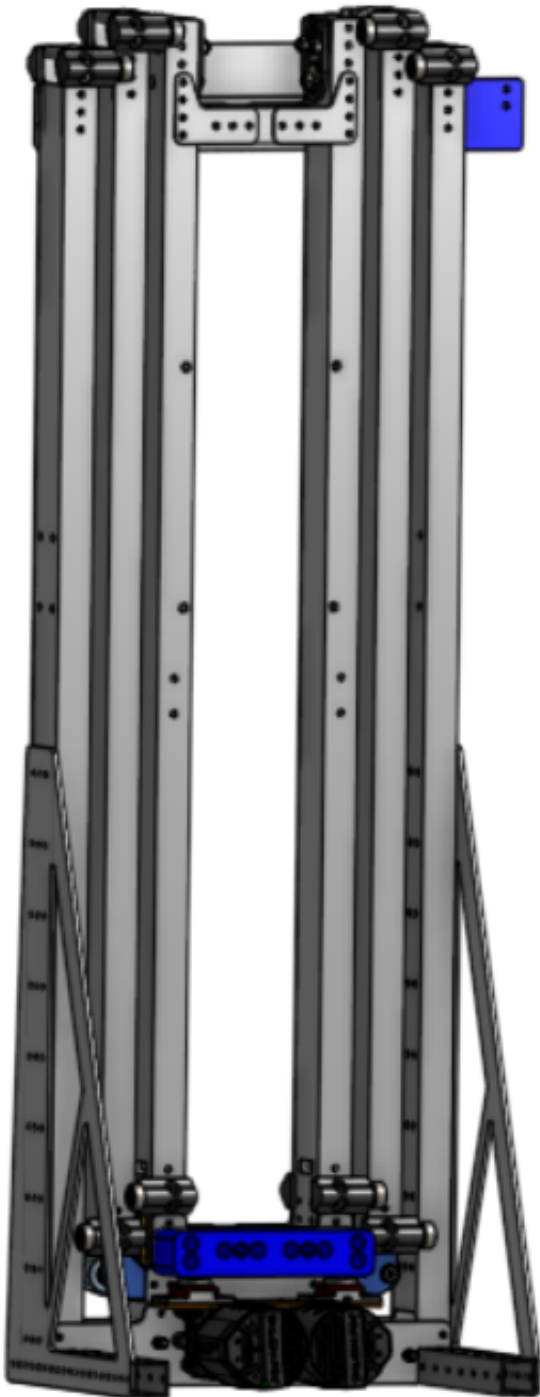
Cage Guide System:

- A ramp prevents cage getting stuck while going over bumpers
- PLA walls bring the cage into the exact position while protecting the other mechanisms

ELEVATOR

2024-2025

DESIGN REVIEWS, ITERATION & FINAL DESIGN



ELEVATOR

The elevator reaches the top branches of the REEF, allowing us to score on all levels. It houses the End Effector as well, which scores CORAL on various branch levels and removes algae from the REEF.

Build:

- Fixed stage is adorned with bling to give robot status to drivers
- Zeroing system using bumper switches
- 2 stage elevator
- 1/16" stock with custom hole pattern
- 6061 aluminum
- Bearing blocks

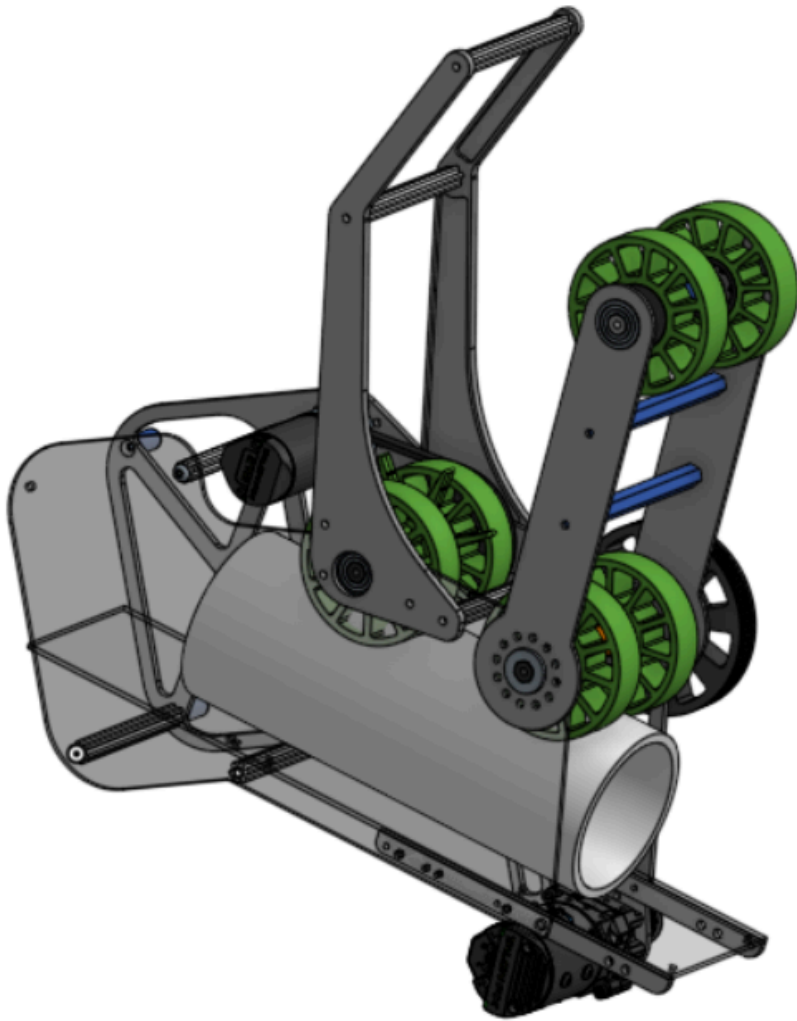
Gearbox:

- 9:1 Versa Planetary
- Two Kraken x60
- Belt run

END EFFECTOR

2024-2025

DESIGN REVIEWS, ITERATION & FINAL DESIGN



END EFFECTOR

Attached to the inside profile of the elevator, the end effector intakes and deposits CORAL on various branch levels of the REEF. The coral piece, taken from the REEF STATION, is transferred from the funnel to the end effector through the wheels. When it's ready to deposit and score, the CORAL is transferred from the top the end effector to the extended part, which then fires onto the branch.

Additionally, there is an “algae claw” with wheels that help remove the algae from the REEF and score it in the BARGE NET.

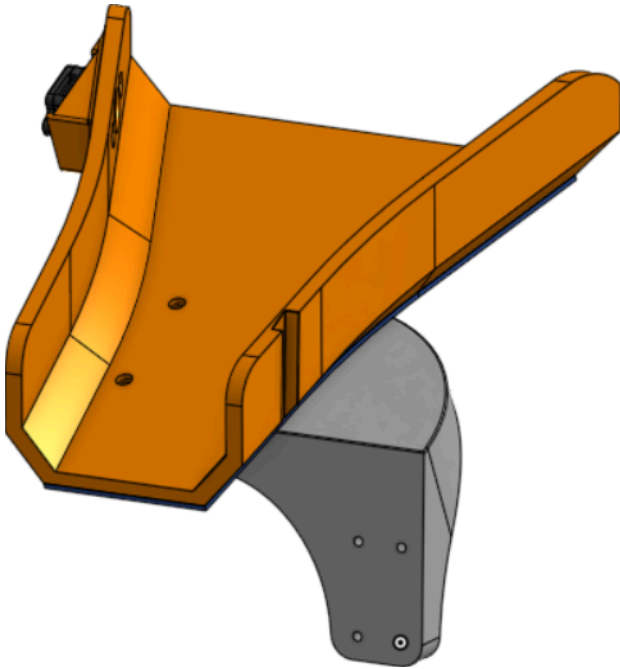
Pulley System:

- X44 Kraken, located inside the end effector for space efficiency
- 56 tooth HTD pulley
- 55 tooth 5 mm belt
- 4” compliant wheel x4
- 4” compliant star wheel

Tongue (Bottom Plate):

- Back plate is at a 30° angle to receive coral from funnel
- Front plate is at a 30° angle for maximum efficient scoring on Branches
- Also extends past original edge of end effector for longer control of coral which contributes to better accuracy

END EFFECTOR (cont.)



Funnel (see left):

- Custom 3D printed 0.40 in. thick PLA
- Mounted at a 30° angle from the ground
- 11 in. wide opening at widest part
- Mounted to stationary level of the elevator
- Has bling for extra visibility

Algae Remover:

- 1/8 lexan “horn” plates catch algae
- Compliant wheels “roll” the algae into grasp
- Connected to coral end effector’s x44 kraken, driven through connected pulleys
- 1/8 lexan “extenders” release to capture algae

FLOOR COLLECTOR

2024-2025

DESIGN



FLOOR COLLECTOR

Otherwise known as the “janitor mechanism” – because it “cleans” up game pieces from the floor – this mechanism picks up algae and coral from the floor to score it in other locations. It was designed to improve our efficiency at scoring in the TROUGH (L1) and in the PROCESSOR. It intakes algae from the floor and holds it close to the drivetrain, while it intakes coral from the floor and holds it in the upper coral plate.

FLOOR COLLECTOR (cont.)

Superstructure:

- $\frac{1}{8}$ aluminum plates, riveted to sides of drivetrain
- Connects to $\frac{3}{8}$ lexan plates on upper half
- Supported by $\frac{3}{8}$ hex axles– act as pivot & structure
- REV Through Bore encoder to track absolute position

Pulley System:

- 25:1 ratio REV max planetary controls pivot movement
- Kraken x44 controls upper half rollers
- 5:1 ratio for rollers

Algae Holding:

- Lower row of 8 compliant wheels intake algae and hold close to drivetrain
- Support axles act as a hard stop, prevents algae from intaking too far

Coral Holding:

- 1/16 lexan sheet curved to match curvature of coral
- 16 compliant wheels hold & deposit coral in upper half
- Support axles also prevent coral from getting stuck in the robot

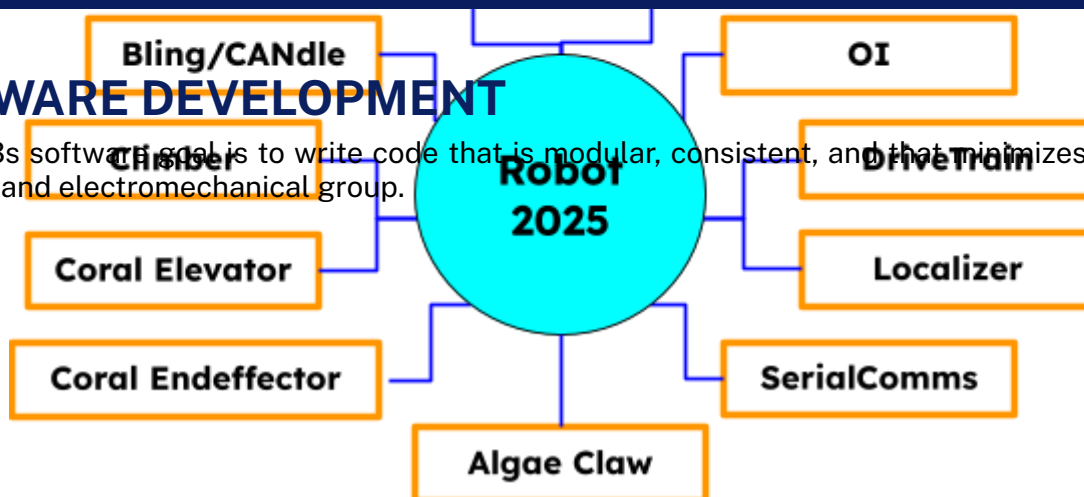
SOFTWARE DEVELOPMENT

2024-2025

SUBSYSTEMS, SENSORS, AUTO & GITHUB

SOFTWARE DEVELOPMENT

Team 1073s software goal is to write code that is modular, consistent, and that minimizes the load on the drivetrain and electromechanical group.



SUBSYSTEMS

Our software is split into subsystems, which usually mirror physical mechanisms on the robot and have their own commands, helper classes, and sensor integration.

This modular separation enables us to work on multiple parts of the robot at once, easily adapt to changes in the physical design of our robot, and reuse the same code in multiple places.

To further allow for reuse of commands, many of our commands take boolean configuration options. A “wait” configuration on several of our basic control commands allows the command to be used both in teleoperated command sequences and in autos; debug options let us bypass the rangefinder and use manual values for testing.

AprilTagFinder

This subsystem uses a system of cameras to collect information about the April Tags on the field. Knowing the position of the tag relative to the robot and the position of the tag on the field, the AprilTagFinder generates robot position measurements that are inputted to the localizer and other subsystems.

Helper Classes

Photon Vision Packages – this package contains classes that fetch and set data from the Photon Vision Interface to help the camera server track April Tags on the field.

FieldMap - a class that contains an AprilTagFieldLayout that stores the translation and rotations of each April Tag on the field.

VisionMeasurement - a wrapper class that stores an April Tag ID, position, time stamp, and range information to easily store information relating April Tags on the field to the robot.

Commands

N/A

Bling/CANdle

This subsystem turns on the light strips on our robot and changes the colors to serve as a diagnostic for the status of other subsystems.

Helper Classes

N/A

Commands

N/A

Climber

This subsystem uses the climber to raise the robot onto the deep cage

Commands

ClimberTeleop – controls the climber motor velocity in Tele-Op

DisengageClimber - moves the climber to the disengaged position

EngageClimber - moves the climber to the engaged position

ZeroClimber - moves the climber to the zero position

Drivetrain

Four SwerveModule objects are created and appropriate configuration information is passed in so the robot knows how they are set up on the actual robot.

Using WPILib's SwerveDriveKinematics, we convert ChassisSpeeds in an (X,Y) coordinate system to individual wheel speeds and rotations. Using the MathUtils class, we are able to optimize the angle to rotate the least in order to get to the desired rotation.

Drivetrain (cont.)

Helper Classes

MathUtils – Custom angle wrapping that puts an angle between $-\pi$ and π or -180° and 180° .

SwerveModule – The code that controls individual swerve modules.

SwerveModuleConfig – The configurations for the swerve modules (constants).

SwerveModuleIDConfig – The configurations for the IDs of the swerve modules.

Commands

TeleopDrive – The command that controls the drivetrain in Teleop. It uses Schema and sends them to the SchemaArbiter.

Elevator

This subsystem uses velocity control to lift the end effector to score or load game pieces.

Helper Classes

N/A

Commands

CoralElevatorTeleop – lifts the elevator using the joystick

CoralElevatorToHeight - lifts the elevator to the desired position, this command takes in a level, corresponding to the branches, and lifts the elevator to that height.

ZeroElevator - moves the elevator to the zero position

End Effector

This subsystem uses the funnel to collect coral from the coral station, transfer them to the end effector, and place coral on the branches or in the trough. It includes a LaserCAN after the rollers to identify collected coral, and another LaserCAN sensor under the end effector to help align to the branches.

Commands

CancelLoadCoral – Sets the end effector motor velocity to zero

CoralEndeffectorTeleop – Ejects a note from the collector.

LoadCoral – Runs end effector motor at the correct velocity to move coral into the mechanism, sets the velocity to zero when the LaserCAN detects a coral.

ScoreCoral – Runs end effector motor at the correct velocity to score coral, sets the velocity to zero when the LaserCAN no longer detects a coral.

AlgaeClawTeleop – Runs the algae rotate motor, moving the pivot between the set minimum and maximum positions.

LoadAlgaeTeleop – Runs algae collect motor with a positive velocity to intake algae into mechanism.

ScoreAlgaeTeleop – Runs algae collect motor with a negative velocity to eject algae from algae claw mechanism

End Effector (cont.)

Commands

LoadAlgaeAuto – Runs the algae rotate motor to the maximum position and runs the collect motor with a positive velocity to collect algae.

ScoreAlgaeAuto – Runs the algae rotate motor to the maximum position and runs the collect motor with a negative velocity to eject algae.

Floor Collector

This subsystem collects coral and algae from the ground and scores in the TROUGH (L1) and NET respectively.

Commands

FloorIntake – Runs the floor pickup collector motor in the clockwise direction to intake coral or algae

FloorEject – Runs the floor pickup collector motor in the counterclockwise direction to eject coral or algae

FloorPickupPivotTeleop – Toggles the floor pickup mechanism between three set heights by running the floor pickup pivot motor

Localizer

Throughout the match, this subsystem corrects the predicted robot position on the field. The subsystem feeds multiple position measurements from AprilTagFinder and odometry into a SwerveDrivePoseEstimator, providing a more robust and accurate measurement of the robot's physical location on the field.

Helper Classes

SwerveDrivePoseEstimator – a position estimator that gradually corrects the robot's estimated position through Kalman filter logic that accepts various robot positions.

Commands

AlignToTag – uses the localizer position and a desired position to align an april tag.

AlignToTagRelative – uses the central relative position of an april tag in relation to the robot along with the localizer position aligned to the reef based on local measurements for accuracy rather than global measurements which are better for robustness.

SmartAlign - A sequential command group that runs *AlignToTag*, *AlignToTagRelative*, and *LidarAlign* to align the robot to the reef

Lidar

This subsystem receives data from the 2D LiDAR Scanner, converts the data received to Scan packets including angle, range, quality, and an x and y value in robot coordinates.

Helper Classes

LidarIO – a thread that uses synchronized methods to read and parse the data received from the LiDAR

Scan - an object that stores the angle, range, quality, and x and y robot coordinates, used to store the data received from the LiDAR

Lidar (cont.)

Commands

LidarAlign – uses the average angle and slope from the Lidar to align to the reef.

OI

Receives data from the Operator Interface (OI) inputs to use in Teleoperated commands

Helper Classes

MathUtil – clamp controller values between custom numbers

Commands

N/A

SENSORS

Our robot's sensors help to both increase our consistency and to provide assistance to our driveteam.

Vision

Our robot uses 4 OpenMV OV9281 cameras to detect april tags, with the end goal of being able to assist the driver in lining up to field elements and to determine the robot's location on the field.

Motor Encoders

All of our motors have built-in encoders for motor control and autonomous positioning.

CANcoders

We use CANcoders on our drive base for motor position calibration.

Limit Switch

We use Limit Switches on the elevator to determine when the elevator has reached it's minimum position and to set the zero position.

LaserCAN

We use the LasrCAN to detect when a coral is in the end effector.

OpenMV Camera

We use the OpenMV camera to detect when the end of the end effector is lined up with the branches.

LiDAR

The LiDAR is mounted on the top of a swerve module, we use the LiDAR to align the robot to the reef when in close proximity to it

AUTOS

Software worked closely with strategy to decide which autos to prioritize. We aimed for autos that could score consistently with minimal interference with other teams.

Pathing

Infrastructure Classes

DrivePath – A class that allows the robot to follow a Path in autonomous mode.

Path – A data structure that organizes a path for the robot to follow. Contains several smaller classes, like Point and Segment. An instance of this class consists of an ArrayList of Segments, and a final orientation for the robot to be in.

Point – A point on the field. Consists of an x and a y coordinate A component for the Path class.

Segment – A segment between two points. It consists of two Points for endpoints, an orientation that the robot aims to reach by the end of the segment, and a velocity with which to traverse the segment.

FieldMap - Provides field-centric positioning for the robot inside and outside of autonomous mode. Allows us to get field-centric coordinates for various AprilTags around the field.

Autos

Our autonomous routines are organized by start location and the “level” of the auto. The position is determined by the orientation of the driver station of whichever alliance our team is on.

Positions:

Left Start - Starts on the left side of the field.

Right Start - Starts on the right side of the field.

Center Left Start - Starts in the center of the field and scores leftwards.

Center Right Start - Starts in the center of the field and scores rightwards.

Levels:

Leave - The robot goes 1 meter forwards, leaving the starting line.

ScoreL<N> - The robot approaches the reef and scores on a specified branch level. If the start position is Left Start or Right Start, it then goes towards the coral station

Score2L4 - The robot approaches the reef, scores on L4, then goes to the coral station to collect another coral, which it then scores on L4.

TESTING AND ANALYSIS

2024-2025

Sharkira

TESTING AND ANALYSIS INTEGRATION

Each mechanism went through a series of tests, iterations, and trials before its current design. See chart below to follow our testing process and mechanism development/integration.

Mechanism	Basic Stage	Control Stage	Tuning Stage	Integrated Stage
Elevator	- Tube stock and bearing blocks	- PID control - Limit switch to find zero position	- PID tuning - Actuation with rolling parallel	- Wheel speed proportional to robot speed
End Effector	- Rolling wheels - Different angle testing	- PID control - LaserCAN sensor to detect coral		- LaserCAN on end to detect branches
Algae Claw	- Basic lexan plate - Two compliant wheels - Knocks algae off the reef using friction			- Lexen plates curved to match the arch of the algae - Collects / intakes the algae using compliant wheels

<i>Climber</i>	- Weight support test (belay system catch*)	- PID control - CANcoder to detect motor position	- Software control - PID	- Tuned controls - Higher speeds - Bling diagnostics
<i>Camera Systems</i>	- REEF detection - CORAL detection - APRILTAG alignment	- Robustness test against exposure	- Calibrate camera - Live video feed on dashboard (raw and processed)	- Aligning to AprilTag centered, to the left, and to the right - Align to the coral stations using AprilTags - continuously update field position
<i>Bling</i>	- Visible from all sides of the robot		- Color sequences/patterns - Usage as diagnostic in mechanisms	- Diagnostic for end effector, battery voltage, branch alignment, and climber position

COMPETITION SEASON DEVELOPMENT

Climber Alterations

We changed our Climber from Version 1 to Version 2 in efforts to improve robot weight, efficiency, and reliability.

Version 1:

- Composed of two parts: Climber Lift and Climber Claw
 - Lift: connected to the claw, pushed the cage down into the center of the robot
 - in order to climb
 - Claw: connected to the lift, gripped the bars of the cage
- Required a lot of torque in order to grab the cage bars and push it down with enough force to push the cage down into the center of the robot
- Weighed ~30 pounds
- Took up a good portion of the robot

Version 2:

- Composed of one part: Climber Claws
 - Claws: push the bottom of the cage down into the robot
- Requires less torque and force to push the robot down
- Weighs ~10 pounds
- Takes up significantly less space
- More efficient & reliable

End Effector Alterations

We modified and added to our End Effector to improve efficiency, accuracy, and reliability. Additionally, our algae removing mechanism went through several iterations to improve its overall scoring capability.

Version 1:

- Two plates driven by a set of wheels on a rolling shaft
- Had a fixed funnel to passively pass coral from human player station to end effector

- Needed a way to remove algae from reef
- Couldn't reliably put coral on reef branches

Extended Tongue:

- Provides more control and guidance over the coral game piece
 - Previously: required the elevator to go up a significant distance in order to score on a specific branch, not always reliable
 - Now: requires the elevator to go above the branch slightly, but not as much as before, to score on a specific branch, very reliable if aligned properly

Algae Removing Mechanism:

Version 1:

- Two lexan plates with wheels driven by the current x44 Kraken motor
- Was made short to stay within frame perimeter
- Uses friction to drive off the front roller of end effector

Version 1 Observations:

- Only able to remove algae from the reef, not able to score in PROCESSOR or NET
- Requires precise movement, tedious
- Passive control resulted in it falling over and messing up coral scoring

Version 2:

- New lexan plates curved to algae arch
- Grips the algae and holds until depositing
- Kraken x60 motor for pivot
- Kraken x44 motor for wheel control

Version 2 Observations:

- Not efficient, took too long to grab algae from the reef
- Did not have stable grip on the algae, difficult to keep hold of it to then score in the NET
- Tedious and requires extremely precise control to properly grab

Version 3:

- 1/8 in. lexan "horn" plates catch algae
 - Compliant wheels roll algae into "horn" grasp
 - Powered by end effector x44 motor through pulley system
 - Requires less precise control, more forgiving
-

SWOT ANALYSIS

Strengths:

- Predetermined design and iteration processes
- Vision processing
- Strong Software and Electromechanical members
- PID tuning and mechanism integration
- Mentor SMEs, including alumni
- Hollis Brookline Robotics Booster club support
- Digitized technical and planning communication (team email, SportsYou)
- Pre-scouting and data gathering
- Continuing developmental processes throughout season
- 3 FTC teams give us a stronger pipeline

Weaknesses:

- Even distribution of work
- Built in software testing time
- Time management
- Mill and CNC training
- Number of graduating students
- Limited storage for prototype/raw materials
- Knowledge transfer gap
- Limited access to workspace (access determined by only 1 person)

Opportunities:

- Connecting with new sponsors
- Student access to new technology
- Learning about integration testing
- Learning about verifications and requirements
- Applying data-gathering practices from 2024 season
- Working with outside groups and teams for mechanism fabrication
- Offseason project planning
- Blending project management styles (SCRUM, Gantt Chart, Team Calendar)
- Technical work with FTC students

- Event volunteering across all programs (familiarity with event proceedings)

Threats:

- Monetary constraints
- Low fundraising opportunities
- Available hours
- Facility policies